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Practical-2:- perfrom data inspection,cleanning,handling,missing value duplicate rewmoval and data transromation using pandas

Step-1 create a data frame

```
#Create student data
import pandas as pd
data = {
    "ID": [101,102,103,104,104,105,106,107],
    "Name": ["Rahul","priya","Amit",None,"Riya","Ajay","Naresh","jenil"],
    "Age": [20,21,None,20,20,22,23,28],
    "city" : ["Surat","Navsari","Surat","Bharuch","Bharuch","Navsari","Surat","Ahmedabad"],
    "Marks":[85,None,78,88,88,81,90,93]
}
df = pd.DataFrame(data)
df
```

	ID	Name	Age	city	Marks
0	101	Rahul	20.0	Surat	85.0
1	102	priya	21.0	Navsari	NaN
2	103	Amit	NaN	Surat	78.0
3	104	None	20.0	Bharuch	88.0
4	104	Riya	20.0	Bharuch	88.0
5	105	Ajay	22.0	Navsari	81.0
6	106	Naresh	23.0	Surat	90.0
7	107	jenil	28.0	Ahmedabad	93.0

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Step-2 inspect the Dataset

```
#Show first 5 rows
df.head()
```

	ID	Name	Age	city	Marks
0	101	Rahul	20.0	Surat	85.0
1	102	priya	21.0	Navsari	NaN
2	103	Amit	NaN	Surat	78.0
3	104	None	20.0	Bharuch	88.0
4	104	Riya	20.0	Bharuch	88.0

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```
# Display last 5 rows
df.tail()
```

	ID	Name	Age	city	Marks
3	104	None	20.0	Bharuch	88.0
4	104	Riya	20.0	Bharuch	88.0
5	105	Ajay	22.0	Navsari	81.0
6	106	Naresh	23.0	Surat	90.0
7	107	jenil	28.0	Ahmedabad	93.0

```
# Display information about the dataset
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8 entries, 0 to 7
Data columns (total 5 columns):
 #   Column  Non-Null Count  Dtype
```

```
---  ------  --------------  -----
 0   ID      8 non-null      int64
 1   Name    7 non-null      object
 2   Age     7 non-null      float64
 3   city    8 non-null      object
 4   Marks   7 non-null      float64
dtypes: float64(2), int64(1), object(2)
memory usage: 452.0+ bytes
```

```
#Show basic statistical information
df.describe()
```

	ID	Age	Marks
count	8.00	7.000000	7.000000
mean	104.00	22.000000	86.142857
std	2.00	2.886751	5.209881
min	101.00	20.000000	78.000000
25%	102.75	20.000000	83.000000
50%	104.00	21.000000	88.000000
75%	105.25	22.500000	89.000000
max	107.00	28.000000	93.000000

```
# Display dimension of dataset
df.shape
```

```
(8, 5)
```

```
# Display name of column.
df.columns
```

```
Index(['ID', 'Name', 'Age', 'city', 'Marks'], dtype='object')
```

```
# Display datatype of each column.
df.dtypes
```

	0
ID	int64
Name	object
Age	float64
city	object
Marks	float64

dtype: object

```
# Check for Missing values.
df.isnull()
```

	ID	Name	Age	city	Marks
0	False	False	False	False	False
1	False	False	False	False	True
2	False	False	True	False	False
3	False	True	False	False	False
4	False	False	False	False	False
5	False	False	False	False	False
6	False	False	False	False	False
7	False	False	False	False	False

```
#print the count of null value for each
df.isnull().sum()
```

	0
ID	0
Name	1
Age	1
city	0
Marks	1

dtype: int64

```
#print the total number of null valaue for whole data set
df.isnull().sum().sum()
```

```
np.int64(3)
```

Step-5 fill missing value

```
# Fill missing Marks values with the avg Marks
df["Marks"] = df["Marks"].fillna(df["Marks"].mean())
df
```

	ID	Name	Age	city	Marks
0	101	Rahul	20.0	Surat	85.000000
1	102	priya	21.0	Navsari	86.142857
2	103	Amit	NaN	Surat	78.000000
3	104	None	20.0	Bharuch	88.000000
4	104	Riya	20.0	Bharuch	88.000000
5	105	Ajay	22.0	Navsari	81.000000
6	106	Naresh	23.0	Surat	90.000000
7	107	jenil	28.0	Ahmedabad	93.000000

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```
# Fill missing Age values with the avg Age
df["Age"] = df["Age"].fillna(df["Age"].mean())
df
```

	ID	Name	Age	city	Marks
0	101	Rahul	20.0	Surat	85.000000
1	102	priya	21.0	Navsari	86.142857
2	103	Amit	22.0	Surat	78.000000
3	104	None	20.0	Bharuch	88.000000
4	104	Riya	20.0	Bharuch	88.000000
5	105	Ajay	22.0	Navsari	81.000000
6	106	Naresh	23.0	Surat	90.000000
7	107	jenil	28.0	Ahmedabad	93.000000

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```
#Fill missing values in the Name column with "Unknown"
df["Name"] = df["Name"].fillna("Unknown")
df
```

	ID	Name	Age	city	Marks
0	101	Rahul	20.0	Surat	85.000000
1	102	priya	21.0	Navsari	86.142857
2	103	Amit	22.0	Surat	78.000000
3	104	Unknown	20.0	Bharuch	88.000000
4	104	Riya	20.0	Bharuch	88.000000
5	105	Ajay	22.0	Navsari	81.000000
6	106	Naresh	23.0	Surat	90.000000
7	107	jenil	28.0	Ahmedabad	93.000000

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Step-6 Detect Duplicate records

```
#detect the duplicate values
df.duplicated()
```

	0
0	False
1	False
2	False
3	False
4	False
5	False
6	False
7	False

dtype: bool

```
# Find total duplicate rows
df.duplicated().sum()
```

```
np.int64(0)
```

Step-7 Remove the duplicated records

```
# Remove duplicate rows
df.drop_duplicates()
```

	ID	Name	Age	city	Marks
0	101	Rahul	20.0	Surat	85.000000
1	102	priya	21.0	Navsari	86.142857
2	103	Amit	22.0	Surat	78.000000
3	104	Unknown	20.0	Bharuch	88.000000
4	104	Riya	20.0	Bharuch	88.000000
5	105	Ajay	22.0	Navsari	81.000000
6	106	Naresh	23.0	Surat	90.000000
7	107	jenil	28.0	Ahmedabad	93.000000

Step-8 Rename a column

```
# Rename the "Marks" column to "Score"
df.rename(columns={"Marks":"Score"},inplace=True)
df
```

	ID	Name	Age	city	Score
0	101	Rahul	20.0	Surat	85.000000
1	102	priya	21.0	Navsari	86.142857
2	103	Amit	22.0	Surat	78.000000
3	104	Unknown	20.0	Bharuch	88.000000
4	104	Riya	20.0	Bharuch	88.000000
5	105	Ajay	22.0	Navsari	81.000000
6	106	Naresh	23.0	Surat	90.000000
7	107	jenil	28.0	Ahmedabad	93.000000

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Step-9 change data type

```
# Convert the Age column values into integer data type
df["Age"]=df["Age"].astype(int)
df
```

	ID	Name	Age	city	Score
0	101	Rahul	20	Surat	85.000000
1	102	priya	21	Navsari	86.142857
2	103	Amit	22	Surat	78.000000
3	104	Unknown	20	Bharuch	88.000000
4	104	Riya	20	Bharuch	88.000000
5	105	Ajay	22	Navsari	81.000000
6	106	Naresh	23	Surat	90.000000
7	107	jenil	28	Ahmedabad	93.000000

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Step-10 Create a new colums

```
#create a new column
df["Final_Score"] = df["Score"] + 5
df
```

	ID	Name	Age	city	Score	Final_Score
0	101	Rahul	20	Surat	85.000000	90.000000
1	102	priya	21	Navsari	86.142857	91.142857
2	103	Amit	22	Surat	78.000000	83.000000
3	104	Unknown	20	Bharuch	88.000000	93.000000
4	104	Riya	20	Bharuch	88.000000	93.000000
5	105	Ajay	22	Navsari	81.000000	86.000000
6	106	Naresh	23	Surat	90.000000	95.000000
7	107	jenil	28	Ahmedabad	93.000000	98.000000

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Step-11 Convert text in uppercase

```
#convert into uppercase
df["Name"] = df["Name"].str.upper()
df
```

	ID	Name	Age	city	Score	Final_Score
0	101	RAHUL	20	Surat	85.000000	90.000000
1	102	PRIYA	21	Navsari	86.142857	91.142857
2	103	AMIT	22	Surat	78.000000	83.000000
3	104	UNKNOWN	20	Bharuch	88.000000	93.000000
4	104	RIYA	20	Bharuch	88.000000	93.000000
5	105	AJAY	22	Navsari	81.000000	86.000000
6	106	NARESH	23	Surat	90.000000	95.000000
7	107	JENIL	28	Ahmedabad	93.000000	98.000000

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Step-12 Convert into lowercase

```
#convert into lowercase
df["city"] = df["city"].str.lower()
df
```

	ID	Name	Age	city	Score	Final_Score
0	101	RAHUL	20	surat	85.000000	90.000000
1	102	PRIYA	21	navsari	86.142857	91.142857
2	103	AMIT	22	surat	78.000000	83.000000
3	104	UNKNOWN	20	bharuch	88.000000	93.000000
4	104	RIYA	20	bharuch	88.000000	93.000000
5	105	AJAY	22	navsari	81.000000	86.000000
6	106	NARESH	23	surat	90.000000	95.000000
7	107	JENIL	28	ahmedabad	93.000000	98.000000

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Step-13 Remove extra space

```
# Remove the extra space
df["city"] = df["city"].str.strip()
df
```

	ID	Name	Age	city	Score	Final_Score
0	101	RAHUL	20	surat	85.000000	90.000000
1	102	PRIYA	21	navsari	86.142857	91.142857
2	103	AMIT	22	surat	78.000000	83.000000
3	104	UNKNOWN	20	bharuch	88.000000	93.000000
4	104	RIYA	20	bharuch	88.000000	93.000000
5	105	AJAY	22	navsari	81.000000	86.000000
6	106	NARESH	23	surat	90.000000	95.000000
7	107	JENIL	28	ahmedabad	93.000000	98.000000

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Step=14 Sort the dataset

```
#Sort the dataset value
df.sort_values(by=['Final_Score'], inplace=True)
df
```

	ID	Name	Age	city	Score	Final_Score
2	103	AMIT	22	surat	78.000000	83.000000
5	105	AJAY	22	navsari	81.000000	86.000000
0	101	RAHUL	20	surat	85.000000	90.000000
1	102	PRIYA	21	navsari	86.142857	91.142857
3	104	UNKNOWN	20	bharuch	88.000000	93.000000
4	104	RIYA	20	bharuch	88.000000	93.000000
6	106	NARESH	23	surat	90.000000	95.000000
7	107	JENIL	28	ahmedabad	93.000000	98.000000

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Step=15 Save the clean data

```
#save Clean Dataset
df.to_csv("Cleaned_Students.csv",index=False)
df
```

	ID	Name	Age	city	Score	Final_Score
2	103	AMIT	22	surat	78.000000	83.000000
5	105	AJAY	22	navsari	81.000000	86.000000
0	101	RAHUL	20	surat	85.000000	90.000000
1	102	PRIYA	21	navsari	86.142857	91.142857
3	104	UNKNOWN	20	bharuch	88.000000	93.000000
4	104	RIYA	20	bharuch	88.000000	93.000000
6	106	NARESH	23	surat	90.000000	95.000000